Model Monitoring & Data Drift

- Machine learning creates static models from historical data. But, once deployed in production, ML models become unreliable and obsolete and degrade with time.

- The reasons could be
    a. Changes in the data distribution in production, thus causing biased predictions.
    b. User behavior itself might have changed compared to the baseline data the model was trained on.

- Both the reasons above are types of Data drift, where first one is a covariate drift and the second one is concept drift.

- Model monitoring is the process of continuously flagging such drifts and automating jobs for retraining the model with new data to ensures that the model remains relevant in production and gives fair and unbiased predictions over time.

References

Types of Data Drift #1

- **Concept Drift**: The statistical properties of the target variable, which the model is trying to predict, changes over time. This causes problems because the predictions become less accurate and become unreliable.

- Examples
    a. COVID-19 brought abrupt changes in consumer behavior and impacted the accuracy of forecasting models.
    b. UPI (Unified Payment Interface) abrupted consumer's spending method, brought many of them online.

- One of the main reasons for concept drift to occur is the non-stationarity of data i.e., change in statistical properties of data with time.

Types of Data Drift #2

- **Covariate Drift**: Covariate shift is the change in the distribution of one or more of the independent variables or input variables of the dataset. This means that the distribution of the feature itself has changed.

- When statistical properties of this input data change, the same model which has been built before will not provide unbiased results. This leads to inaccurate predictions.

- Examples
    a. Suppose a model is trained with an income variable and that is in production. Over time, if income of the population increases by 5-10%, the model encounters real-time data drift with increased income. The model sees an increase in mean and variance, and therefore it leads to a data drift.

    a. There is a drift in Marital status of the population.

Methods for Detecting Data Drift

- All the methods for detecting data drift are lagging indicators of drift i.e. Only after they have processed enough data after any kind of drift that has occurred, that the actual drift is detected.

- **Kolmogorov-Smirnov (K-S) test**:
    a. The K-S test is a nonparametric test that compares the cumulative distributions of two data sets, the training data and the post-training data.
    b. The null hypothesis is that the two distributions are identical and the alternative is that they are not identical. If the null is rejected then we can conclude that there is a drift in the model.
    c. The test is valid for numerical columns.

- **Chi-squared test**:
    a. The chi-squared two-sample test is applied to the categorical features to identify data drift.

Population Stability Index

- **Population Stability Index**

    a. PSI was originally developed in the Banking and Finance industries for testing the changes in the distribution of a risk score over time.

    a. It’s being used both for detecting numerical and categorical variables data-drifts. *How for categorical variables?* Dividing the data into buckets.

    a. KL(Kullback-Leibler) divergence is a good measure to find how much a observed distribution ‘*o*’ differs from it's ideal/reference ‘*r*’ version.

    $$\sum_i r_i * \log\left(\frac{r_i}{o_i}\right)$$

    a. KL divergence is not symmetrical though, i.e. if we permute '*o*' and '*r*', we won’t necessarily find the same value.

Population Stability Index

- To address this issue, one can use a symmetrical version of the KL divergence, it's called the Jeffreys divergence, often known as the Population stability index (PSI). It's defined as the sum of KL divergence from '*o*' to '*r*' and the one from '*r*' to '*o*'.

$$\sum_i (r_i - o_i) * \log\left(\frac{r_i}{o_i}\right)$$

- When PSI<=0.1
  - This means there is no change or shift in the distributions of both datasets.

- When 0.1<PSI<0.2
  - This indicates a slight change or shift has occurred.

- When PSI>0.2
  - This indicates a large shift in the distribution has occurred between both datasets.

Model-Based Approach

- A Machine Learning model-based approach can also be used to detect data drift between two populations.

- Idea is to label the data which has been used to build the current model in production as 0 and the real-time data gets labeled as 1.

- If the model gives high accuracy, it means that it can easily discriminate between the two sets of data. Thus, we could conclude that a covariate shift has occurred and the model will need to be recalibrated.

- On the other hand, if the model accuracy is around 0.5, it means that it is as good as a random guess. This means that a significant data shift has not occurred and we can continue to use the model.

- The disadvantage of this model is that every time new input data is made available, the training and testing process needs to be repeated which can become computationally expensive.

Adaptive Windowing (ADWIN)

- The Adaptive Windowing (ADWIN) algorithm uses a sliding window approach to detect concept drift. This method is applicable for univariate data.

- Window size is fixed and ADWIN slides the fixed window for detecting any change on the newly arriving data.

- A user-defined threshold is set to trigger a warning that drift is detected.

- If the absolute difference between the two means derived from two sub-windows exceeds the predefined threshold, an alarm is generated.

Page-Hinkley method

- This drift detection method calculates the mean of the observed values and keeps updating the mean as and when new data arrives.

- A drift is detected if the observed mean at some instant is greater than a threshold value lambda.

Handling data drift in production

- In production, there are multiple ways to respond to data drift.

- **Blindly update model**: This is a naïve approach. There is no proactive drift detection. Models are periodically retrained and updated with recent data. Without drift detection in place, it is difficult to estimate the time interval for re-training and model re-deployment.

- **Training with weighted data**: When a new model is trained instead of discarding old training data, use weight inversely proportional to the age of data.

- **Incremental learning**: As new data arrives, the models are continuously re-trained and updated. As a result, the model is always adapting to the changes in the data distribution. This approach will work with machine learning models which allow incremental learning one instance of data at a time.